

CHEZTER P. VARGAS

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PROFESSIONAL SUMMARY

Fresh graduate with a bachelor's degree in biomedical engineering and a passion for embedded systems programming. Experienced in sEMG-based Biosignal processing and embedded systems development, while bringing complementary skills in C/C++ programming. Seeking to contribute to R&D workspaces and healthcare/medical environments within a forward-thinking organization. Open to employer-sponsored visa arrangements, while processing residence visa application.

EDUCATION

Bachelor of Science in Biomedical Engineering

Ateneo de Zamboanga University

Aug 2021 – Dec 2025

Zamboanga City, Philippines

- GPA: 3.28 / 4.00
- Thesis: "eM-Brace: Development of an Autonomous Wrist Splint with an Integrated Massage System for Alleviating Muscle Weakness Related to Carpal Tunnel Syndrome"
- Relevant coursework: Anatomy and Physiology for Engineers, Biomaterials and Biocompatibility, Biomedical Instrumentation, Biomedical Signal Processing, Health Technology Management
- Officer, Biomedical Engineering Society — Head of Creatives (2023–2024)

WORK & INTERNSHIP EXPERIENCE

Biomedical Engineering Intern

Zamboanga City Medical Center

Jun 2024 – Jul 2024

Zamboanga City, Philippines

Equipment Calibration & Maintenance

- Performed calibration and functional testing of clinical equipment using Fluke patient simulators, defibrillator analyzers, and medical gas flow analyzers.
- Disassembled, cleaned, and recalibrated portable biomedical devices including medical gas flow meters and fingertip pulse oximeters, for redeployment to hospital departments.
- Conducted in-situ preventive maintenance of fixed, non-transportable equipment (e.g., patient monitors) directly at ward locations using portable calibration tools.

Clinical Engineering Documentation

- Completed and filed work orders in accordance with ZCMC BME departmental standards, maintaining accurate maintenance histories.
- Performed equipment rounds across hospital departments to identify and schedule equipment due for maintenance per the master list.

Medical Makerspace Rotation

- Completed a dedicated rotation in the ENT Medical Makerspace, gaining exposure to FDM 3D printing, slicer software workflows, and basic CAD design.
- Designed and produced a functional phone stand as a solo CAD exercise; successfully printed three models on department FDM printers.

PROJECTS

eM-Brace | Autonomous CTS Rehabilitation Device

Undergraduate Thesis Project

2024 – 2025

Ateneo de Zamboanga University

Designed and developed an autonomous wrist splint integrating a myofascial release massage system for alleviating muscle weakness associated with Carpal Tunnel Syndrome (CTS).

- EMG electrodes positioned on the Abductor Pollicis Brevis muscle detect wrist rest conditions via real-time signal acquisition, processing, and RMS-based threshold classification; achieved 100% accuracy in muscle state classification during functional testing.
- Integrated Massager module calibrated to deliver 1.36 kPa of therapeutic pressure, activated autonomously only during detected muscle rest to prevent unintended stimulation.
- Auto-Fit Module dynamically adjusts splint tension in response to muscle activity, maintaining consistent wrist support pressure of approximately 3.68 kPa across varying wrist sizes.

- Motion Support Module loosens strap tension during active states, permitting up to 10° flexion and 30° extension to accommodate daily wrist movement.
- Beta tested on 15 participants with induced CTS symptoms; 73% demonstrated grip strength improvement exceeding 2.2 kg, with mean pain scores reducing from 3.27 to 0.33.
- Platform: ESP32 WROOM; biosignal pipeline: analogue signal conditioning, notch filtering, digital reconstruction, and threshold-based classification; evaluated by a licensed physical therapist.

SKILLS

Embedded Systems	ESP32; sEMG signal acquisition and processing; sensor integration
Biomedical / Domain	Calibration and preventive maintenance; Fluke patient simulators; defibrillator analyzers; medical gas flow analyzers; fingertip pulse oximeters
Programming	C/C++ -- Intermediate, HTML/CSS/JavaScript -- Beginner
Tools & Software	Git; VS Code; Arduino IDE; MATLAB; BSL Analysis
Clinical Documentation	Work orders; equipment maintenance records; master equipment lists; BME departmental reporting
Soft Skills	Analytical problem-solving, Teamwork, Adaptability, Structured time management

CERTIFICATIONS & TRAINING

- On-The-Job Training Certificate — Zamboanga City Medical Center (2024)
- PCB Development & Design Workshop — Junior IECEP ADZU Chapter (2023)
- Youth Volunteer Orientation Course — Philippine Red Cross (2019)

HONOURS & AWARDS

- Most Diligent Intern — *Zamboanga City Medical Center* (2024)
- Best in Calibration — *Zamboanga City Medical Center* (2024)
- Best Product Demo Performance — *ENT-HNS Medical Makerspace* (2024)
- Jack of All Trades Award — *ENT-HNS Medical Makerspace* (2024)

VOLUNTEER & EXTRACURRICULAR

Head of Creatives	2023 - 2024
Biomedical Engineering Society	Zamboanga City, Philippines
Public Information Officer	2019 - 2020
Senior Plus Red Cross Youth	Zamboanga City, Philippines
Creatives Team Member	2021 - 2022
SITEAO Academic Organization	Zamboanga City, Philippines

LANGUAGES

English: Fluent	Tagalog: Native	Chavacano (Broken Spanish): Intermediate
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REFERENCES

Available upon request.